Release Plan

Product: StudyBudd MVP

Team: StudyBudd

Release Date: March 10, 2026

(Revision 3.0 - March 10, 2026)

1 High Level Goals

The primary goal of this release (MVP) is to deliver a fully functional, AI-powered study assistant platform. The release enables students to upload course materials and interact with them using Generative AI via a RAG (Retrieval-Augmented Generation) pipeline.

- **Core Utility:** Upload and store PDF lecture notes securely, with full text extraction and embedding generation.
- **AI Interaction:** Ask questions and receive answers grounded in uploaded content via a document-aware RAG pipeline (Supabase vector search).
- **Study Aids:** Automatically generate flashcards and quizzes from user documents, with AI-powered explanations for deeper understanding.
- **Productivity:** Provide a Pomodoro timer for focused study sessions.
- **Accessibility:** Responsive cross-device access and shareable study material links.

2 User Stories Defining the Scope

The following is the breakdown of user stories by Sprint, reflecting what was actually planned, completed, and carried over across all four sprints.

Sprint 1: Foundation & Infrastructure (Jan 20 – Jan 27)

Goal: Establish core infrastructure (Docker, React, FastAPI, Supabase), implement the Pomodoro timer and Authentication UI, and conduct LLM research.

- **{High}** US 1.1: As a student, I want to use a Pomodoro timer so that I can maintain focus during study sessions. **[14 hrs]** — *Completed*
- **{High}** US 1.2: As a developer, I want to set up the project infrastructure and research key technologies (Docker, CI/CD, LLM evaluation, testing setup). **[19 hrs]** — *Completed*

- **{High}** US 1.3: As a student, I want to have a fully accessible website with a login system (Authentication UI + Backend). **[19 hrs]** — *Partially Completed (Frontend only; Backend auth carried over to Sprint 2)*

Sprint 1 Result: 2/3 user stories completed. 40/52 planned hours delivered. Key lesson: Frontend and Backend were developed in isolation, leaving backend logic unfinished. Team adopted "vertical slices" approach for Sprint 2.

Sprint 2: Core RAG & Data Management (Jan 28 – Feb 10)

Goal: Achieve "Core Utility" by resolving Authentication, implementing the PDF upload pipeline, enabling the first RAG-powered Q&A, and storing chat history.

- **{Critical}** US 2.0: As a system, I need working end-to-end Backend Authentication so that user data is secure. (Carry-over from Sprint 1) **[10 hrs]** — *Completed*
- **{High}** US 2.1: As a student, I want to upload documents (PDFs) so that I can use them for AI-assisted studying. **[20 hrs]** — *Partially Completed (Upload/download functional; text extraction & embedding generation carried over)*
- **{High}** US 2.2: As a student, I want to ask questions about my notes so that I can better understand the material (RAG Pipeline). **[20 hrs]** — *Partially Completed (Chat interface with AI streaming built; RAG grounding in user documents carried over)*
- **{Medium}** US 2.3: As a student, I want to view my chat history with the AI so that I can revisit previous Q&A sessions. **[7 hrs]** — *Completed*

Sprint 2 Result: 2/4 fully completed, 2/4 partially completed. 38/57 planned hours delivered (85% rate per team metrics). Additional unplanned work delivered: UI redesign, AI chat streaming with markdown rendering, User Profile page. Tooling migration from Linear to Jira occurred mid-sprint.

Sprint 3: AI Generation Tools (Feb 11 – Feb 24)

Goal: Complete the RAG pipeline (carry-over), and deliver flashcard and quiz generation features.

- **{Critical}** US 3.0: As a student, I want the AI to answer questions grounded in my uploaded documents (RAG Pipeline completion — carry-over from Sprint 2). **[23 hrs]** — *Mostly Completed (Embeddings, vector search, PDF extraction done)*
- **{High}** US 3.1: As a student, I want to generate flashcards from my documents so that I can efficiently memorize key concepts. **[16 hrs]** — *Partially Completed (Frontend UI, database schema, prompt template done; backend endpoint carried over)*
- **{Medium}** US 3.2: As a student, I want to generate quizzes from my notes so that I can test my understanding before exams. **[17 hrs]** — *Partially Completed (Frontend UI, database schema, prompt template done; backend endpoint carried over)*

- **{Low}** US 3.3: As a student, I want to share documents with classmates so that we can collaborate on study materials. **[8 hrs]** — *Partially Completed (Share button UI done, access control done; link generation endpoint in progress)*

Sprint 3 Result: 54/79 planned hours completed (68% rate). Midterm exams significantly reduced development capacity. New tool learning (Together AI API) took longer than estimated. Strong frontend and database foundation laid for Sprint 4 completion.

Sprint 4: MVP Completion & Polish (Feb 25 – Mar 10)

Goal: Close out all carry-over work, deliver a fully functional document-aware study platform, and add responsive design and AI explainer features.

- **{Critical}** US 4.0: Complete RAG Pipeline — finish Agent integration and PDF extraction. (Carry-over from Sprint 3) **[8 hrs]**
- **{High}** US 4.1: Flashcard Generation — finish database tables, prompt template, and `/api/flashcards/generate` endpoint. (Carry-over) **[10 hrs]**
- **{High}** US 4.2: Quiz Generation — finish prompt template and `/api/quizzes/generate` endpoint. (Carry-over) **[9 hrs]**
- **{Medium}** US 4.3: As a student, I want to generate shareable links for my notes so that I can share study materials with classmates. (Carry-over + extended to flashcards/quizzes) **[9 hrs]**
- **{High}** US 4.4: As a student, I want to access the application from any device with a browser so that I can study anytime, anywhere. (New — responsive design) **[12 hrs]**
- **{Medium}** US 4.5: As a user, I want to ask AI questions about my flashcards or quiz so that I can get deeper explanations of concepts I don't understand. (New — AI Explain) **[13 hrs]**

Sprint 4 Plan: 61 total hours (36 carry-over + 25 new). Explicit task dependencies documented. Mid-sprint integration checkpoint scheduled for March 3. "Core before stretch" policy enforced — new features (US 4.4, 4.5) only begin after carry-over work is in progress or done.

3 Sanity Check

- **Capacity Analysis:** The team has 4 members. Based on course expectations and individual schedules, each member can dedicate approximately 10–14 hours per week, yielding a total team capacity of 40–56 hours per one-week sprint and 80–112 hours per two-week sprint. Sprint 1 was a one-week sprint (52 hrs planned); Sprints 2–4 were two-week sprints (57, 79, and 61 hrs planned respectively).
- **Actual Velocity:** Sprint 1 delivered 40/52 hrs; Sprint 2 delivered 38/57 hrs; Sprint 3 delivered 54/79 hrs. Average completion rate across sprints is approximately 75%. The team consistently delivered core infrastructure and frontend on time, with backend AI integration (RAG, generation endpoints) being the primary source of carry-over.

- **Carry-Over Pattern:** The RAG pipeline was the longest-running carry-over item (Sprint 2 → 3 → 4), primarily due to the complexity of integrating Together AI embeddings, and Supabase vector search. By Sprint 4, only 8 hours of RAG work remained, indicating convergence.
- **Risk Mitigation Applied:** Based on retrospectives, Sprint 4 adopted explicit task dependency ordering, a mid-sprint integration checkpoint, daily pulls from main, and a "core before stretch" policy. The original Collaborative Editing feature (US 4.1 in the v2 plan) was dropped entirely in favor of completing the AI generation pipeline, which was the correct prioritization call.
- **Sprint 4 Risk:** At 61 planned hours, Sprint 4 is within capacity but includes 6 user stories. The team's primary risk is integration complexity when connecting the finalized backend endpoints to the already-built frontend components. The mid-sprint checkpoint on March 3 is designed to catch issues early.

4 Product Backlog

Items discussed but not included in the MVP Release:

- **Collaborative Editing:** Real-time Google-Doc-style note editing (originally planned for Sprint 4 in v2, descoped due to carry-over from AI pipeline).
- **Mobile App Interface:** A dedicated native mobile application.
- **Gamification:** Badges and leaderboards for study streaks.
- **Calendar Integration:** Syncing Pomodoro sessions with Google Calendar.
- **Image Recognition:** AI processing of images within uploaded documents (explored during development but not included in MVP scope).

5 Revision History

Version	Date	Description
1.0	Jan 20, 2026	Initial release plan.
2.0	Feb 10, 2026	Updated after Sprint 2; adjusted user story numbering and sprint scope.
3.0	Mar 9, 2026	Final revision.

Last modified: March 10, 2026 (Revision 3.0 — Updated based on Sprint 1–4 Plans & Reports)